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What is lost and gained in NFHS-6

NFHS-6 records gains in child nutrition, maternal care, institutional births and women's Internet use, while also introducing new questions on digital literacy and direct benefit transfers, but the survey's preliminary fact sheet is thinner than the last round, dropping key indicators such as anaemia, mortality, sex ratio at birth

ECONOMIC NOTES

Arreeta Arora

India's latest National Family Health Survey (NFHS-6) records gains in child nutrition, maternal care, institutional births and women's Internet use. But its preliminary fact sheet is also thinner than the last round, with 101 indicators compared with 131 in NFHS-5.

Which indicators were removed from NFHS-6?

Key indicators such as anaemia, mortality, sex ratio at birth, sanitation, and clean cooking fuel have been removed from the survey.

The Union Health Ministry released the fact sheets of NFHS-6 on May 29, covering 2023-24. The survey recorded data from nearly 6.8 lakh households across every State and Union Territory except Manipur.

Preliminary findings report clear gains on several measures, including mothers getting at least four antenatal check-ups, up about seven percentage points from NFHS-5, an increase in institutional births and women's Internet usage. It also points to declines in several metrics, such as exclusive breastfeeding of infants under six months, down nearly eight percentage points, and the use of modern contraception, down to 52.7% from 56.4%.

The NFHS is commissioned by the Ministry of Health and Family Welfare, which designates the International Institute for Population Sciences (IIPS) to conduct the survey. Over the years, the scope of the survey has been additive by design, retaining the previous questionnaire and adding to it.

NFHS-4 in 2015-16 introduced district-level estimates and tablet-based digital interviewing for survey collection. NFHS-5 pushed the indicators further, including new topics, such as preschool education, disability, access to a toilet facility, debt registration, bathing practices, digital measurement methods and reasons for abortion. It also extended blood pressure and blood sugar measurements from adults aged between 15 and 49 to all adults aged 15 and above. The survey measured 131 key indicators, up from 114 in NFHS-5.

While the HIV testing component was dropped from NFHS-5, it retained questions on HIV/AIDS knowledge, attitudes, prior testing, sexually transmitted infections, and sexual behaviour. In NFHS-6, biological HIV testing has been brought back as part of the clinical, anthropometric, and biochemical testing schedule.

The NFHS-6 fact sheet does not separately spell out whether all HIV/AIDS knowledge and attitude questions were retained.

NFHS-6 also added new questions on direct benefit transfers, self-help group memberships, digital literacy and financial transactions. It also included testing for Hepatitis-B and Hepatitis-C among women and men, as well as dried blood spot (collection from children aged 4-6) for hepatitis-B testing.

But for the first time, the survey has also subtracted overall, showing a net reduction of 30 indicators in the preliminary results. Among the dropped indicators, the most noticeable were anaemia, infant and child mortality, sex ratio at birth, clean cooking fuel use, and sanitation, have all appeared since at least NFHS-4.

THE GIST

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NFHS-6 shows a net reduction of 30 indicators in its preliminary fact sheet, with 43 indicators dropped and 13 added compared with NFHS-5.

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Anaemia was dropped from NFHS-6, with the official position being that it will be tracked separately through the Diet and Biomarkers Survey.

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Several long-running indicators, including infant and child mortality, sex ratio at birth, sanitation coverage, clean cooking fuel use and cancer-screening measures, are absent from the latest fact sheet, leaving gaps that no other single source fills at the same scale.



The NFHS is commissioned by the Ministry of Health and Family Welfare, which designates the IIPS to conduct the survey. GETTY IMAGES

Why was anaemia dropped?

The case for removing anaemia is related to how it was measured. The indicator had long shown a worsening picture.

Between NFHS-4 in 2015-16 and NFHS-5 in 2019-21, anaemia rose across the board. Among children, anaemia prevalence went up from 58.0% to 67.1%, among women aged 15-49, it rose from 53.1% to 57%, and among pregnant women, anaemia rose from 51.4% to 52.2%.

The rise in anaemia was near-universal across the country, with child anaemia increasing in 28 States and Union Territories, and in some cases by big leaps, from 35.7% to 68.4% in Assam and 39.3% to 46.4% in Mizoram. Such deterioration was reported despite the government launching the Anaemia Mukt Bharat campaign in 2018, aimed at tackling anaemia. The reason for dropping it as an indicator then boiled down to how the data were being collected.

NFHS measured haemoglobin from a finger prick blood sample read on a portable analyser, which several nutrition researchers contended overated anaemia compared to the venous blood drawn by other surveys. IIPS dropped the anaemia questionnaire when NFHS-6 fieldwork began in 2022. The official position is that the condition will now be tracked separately, through a dedicated Diet and Biomarkers Survey under the National Institute of Nutrition using a method its proponents consider more accurate.

The Diet and Biomarkers Survey in India was not a hurried replacement and was launched in December 2022, before NFHS-6 fieldwork began, at the ICMR National Institute of Nutrition in Hyderabad.

The survey recorded data on individual

dietary intake across age groups, paired with blood and urine biomarkers.

It also built upon nutrition deficiencies and tracked obesity alongside anaemia, a first, according to the organizers. For anaemia, data were collected from venous blood instead of the finger prick method that NFHS used. Data collection is complete, but hasn't been released yet.

What other changes were made?

A line-by-line comparison of the two fact sheets from NFHS-5 and NFHS-6 shows that the net fall of 30 actually combines 43 indicators dropped and 13 added. Several of the deletions were long-running series, and a few are closely related to the government's signature programmes.

Prime Minister Narendra Modi's government in 2019 had announced that India would be open defecation free. NFHS-5 recorded 70% of the country's population living in households with access to sanitation facilities. That data point has also been dropped.

The share of households using clean fuel for cooking, 58.0% in NFHS-5, is gone — a direct measure of the success of the Pradhan Mantri Ujjwala Yojana.

The three mortality indicators — neonatal, infant and under-five — have also been cut, but these will be tracked by the Sample Registration System, whose latest bulletin put infant mortality at 24 per 1,000 live births.

The Registration System, however, does not carry district-level data and socio-economic breakdowns that are available in NFHS.

The sex ratio of the total population and the sex ratio at birth, 929 females per 1,000 males in NFHS-5, are both absent, removing a standard signal of sex-selective practices. Four

cancer-screening indicators, covering cervical, breast and oral cancer,

introduced only in NFHS-5 are gone after a single round.

A few changes are redefinitions rather than cuts. Women's individual ownership of a house or land has become a household-level measure. Three dose hepatitis B line has become a birth dose measure, and pre-school attendance has shifted age bands, targeting a younger demographic. IIPS has not published a rationale for the wider list.

Together, the removals leave no current survey-based national figure for infant mortality, sanitation coverage, sex ratio at birth, cancer screening rates or comprehensive HIV knowledge, gaps that no other single-source fills at the same scale.

How did survey results change between NFHS-5 and NFHS-6?

NFHS-6 reported a drop in the number of women who have experienced spousal violence, down to 22.3% from 29.3%. The number of children aged five or below who are stunted declined too. The drop from NFHS-5 to NFHS-6 was just under three percentage points, but NFHS-6 saw a drop of over six percentage points.

The sex ratio of the total population and the sex ratio at birth, 929 females per 1,000 males in NFHS-5, are both absent, removing a standard signal of sex-selective practices. Four

Health insurance coverage rose the most in West Bengal, from 33.7% of households in NFHS-5 to 88.2% in NFHS-6. Women's Internet use saw its largest increase in Andhra Pradesh, from 22% to 63.6%. Haryana recorded the steepest fall in exclusive breastfeeding among infants under six months, from 69.9% to 41.2%. The share of women classified as overweight or obese increased in every State.



Mustard is the most vital crop to meet India's demand of edible oils. (IISc photo)

New mustard hybrids need diverse farming plans

Yasodeen Mahant

In the 2020-21 sowing season, from October to April, Indian farmers are set to begin the wide-scale cultivation of bioherbicide-resistant (Bt-resistant) mustard hybrids. These hybrids are designed to overcome a parasitic weed called Orobanche, a major hurdle in oilseed production.

India has long grappled with an edible oil deficit. In 2019-20, oil country imported around 16 million tonnes at \$18 per 100-lb bag. Mustard is the most vital oilseed crop to meet this demand, especially in the arid environments of North India. However, its yields have been depressed throughout the year by Orobanche, a destructive root parasite that attaches to the mustard plant and drains it of water and nutrients. It has been recently found in central Punjab (IISc).

The new Bt-resistant hybrids are a collection of bioherbicide-resistant (Bt-resistant) mustard (Bt-crops), these hybrids were developed in a process called mutation breeding – where scientists breed crops to possess certain natural mutations. The process is based on an enzyme called acetylcholine synthase (ACh), which is essential for plant growth. Normal mustard plants are killed by Bt herbicides because the herbicides inhibit ACh.

In the new Bt-resistant hybrids, a single change in the plant's DNA rendered the ACh enzyme immune to the herbicide, freeing farmers to spray Bt herbicides directly over the field, killing only the weeds. This strategy is particularly effective against Orobanche because, as the herbicide moves through the soil and the plant, it kills the parasite where manual weeding can't reach.

Farmers have welcomed these hybrids since they don't increase the demand for labour exactly when it might be available. However, in the mustard growing window when weeding is necessary, and manual weeding is becoming increasingly scarce, it protects oilseeds.

Some experts have urged caution over their long-term use. In a 2019 paper published in the journal *Current Science*, Deepak Pental, a leading expert on herbicide resistance at the University of Delhi, acknowledged the transference benefits but also said the technology must be managed with care to prevent the weeds evolving their own resistance.

Prof. Pental explained that if farmers rely on only one type of herbicide year after year, the "wrong directional selection" will lead to the "emergence of resistant or less susceptible weed populations," the erosion of herbicide efficacy, and ultimately the strategic failure of the production system built around that chemical dependency.

To that end, he wrote that the introduction of these crops could not be viewed as a simple trade-off, where chemicals merely replace manual labour, but as a "challenge" that requires a variety of weed control methods.

Specifically, for the technology to remain viable for several more years, it must be part of a broader, more diverse weed management strategy that includes weeding, crop rotation, and other methods. It should be used in conjunction with other types of herbicides and manual weeding, and continuing to use manual weeding to remove any weeds that arise.

Chemical resistance is not the chemical solution. "A single herbicide mode of action is not the sustainable weed management strategy in any agro-ecosystem," he wrote. "In India, rely on the chemical-resistant mustard this season, the new challenge for agro-ecosystem ecology will be to ensure these powerful new tools become part of a diverse, and multifaceted weed management strategy rather than quick fix of chemical resistance."

Amplifying random numbers brings a breakthrough in digital security

Creating a truly unguessable password is difficult as even the best computers know behind subtle, predictable patterns. By tapping into the power of quantum physics, scientists have now found a way to turn mildly random data into flawless security keys that are practically impossible to predict.

Yasodeen Mahant
Orobanche

Most digital security relies today on random numbers to generate cryptographic keys. Think of a cryptographic key like a coin. Complex patterns, if generated is truly random, an attacker has no way to guess every possible combination. If the "random" process used to make that generator has a pattern, the attacker can use that pattern to skip billions of guesses. Digital security firms like Skychip build the numbers it uses for encryption are truly at random as they seem, but in a study published in Nature, researchers at ETH Zurich have demonstrated a new solution: called randomness amplification, where they used quantum physics to turn predictable bits of data into certified perfect randomness.

I saw from the university said the researchers had effectively "generated certified perfect randomness for the first time."

Creating true randomness is actually really difficult. Most random number generators (RNGs) produce bits with a small bias. Say you're flipping a coin that lands on heads 5% of the time instead of 50%. That 5% difference is a bias.

Santha Vazirani limit

In 1986, computer science researchers Mikko Santha and Umesh Vazirani showed that a classical computer cannot upgrade a weakly random source into a perfectly random one. For another way, if a source of random numbers is predictable even in a small way, classical post-processing alone could never eliminate it. So if you give a classical computer a biased coin, any data it produces based on that coin will remain just as biased and predictable, no matter how much the computer tries to fix it on the input. This is why even high-end random number generators suffer from experimental imperfections – the best of them are still biased, and thus, technically, are not random.

"Even modern random number generators, which are based on quantum mechanical effects like the reflection of photons from beam splitters, are not entirely immune to such a systematic error or bias," Andrea Mulm, one of the study's lead investigators and an assistant professor at ETH Zurich, said in the note.

Over the last two decades, theoretical physicists have suggested that quantum physics may provide a workaround. In fact, in 2012, Roger Colbeck and Renato Renner who is one of the authors of the new study – showed in theory that quantum physics could do it.

The ETH Zurich team achieved it using a bit test, a physics experiment designed to prove a phenomenon called quantum entanglement. Entanglement is a state where two quantum particles – like atoms or photons – become linked such that a measurement performed on one particle instantly affects the other, no matter how far apart they are.

In a bit test, two entangled particles are separated and measured. The particles' answers in a particular highly correlated way, it will show that the particles were not secretly decided in advance. In other words, the test will prove the universe itself did not know the answer and the measured the measurement happened.

Now, the quantum physics, the act of measuring something creates new information that wasn't there before. For example, if the two particles will be in a superposition of multiple states. When you measure one, it will collapse into one particular state, and so will the other entangled particle. This "bit test" state is the new information.

Because this information is born at the exact instant of measurement, it can't be possible for anyone to have known the answer in advance. In the team's latest bit test, 2.6 billion bits of information from a single source of randomness.

Over time, the team members entangled two particles, then placed them 30 m apart. This distance made sure the particles could not "cheat" by communicating their "bit" state to each other at the speed of light.

Next, they used random biased bits to decide how to measure each particle. Then, with the measurement outcomes, they calculated the Bell violation score – a measure of how wrong the particles were entangled, and thus how chaotic the universe might have been about the measurement outcomes. The score was 2.23.



Digital security has an Achilles heel: the user's choice for encryption are only as random as they seem. Most security uses

In quantum physics, the act of measuring something creates new information that wasn't there before, because this information is born at the exact instant of measurement. It is impossible for anyone to have known the outcome in advance, becoming an additional source of randomness.

"My understanding is that the main advance here is not device-independent random number generation per se. That has already been demonstrated using Bell tests," Umesh Vazirani, head of the Quantum Information and Computing lab at the Swiss Research Institute, Bangalore, told the Hindu.

The new element in the experimental demonstration of device-independent randomness amplification – starting with a weak, imperfect source of randomness and using quantum correlations to certify output bits that are unbiased under the stated model.

This is significant because practical quantum random number generators are never ideal, and in our own work on QKD and semi-device-independent certification, we have also emphasised that the real issue is not whether a bit string passes statistical tests but whether its unpredictability can be certified from clearly stated physical assumptions," he wrote.

"The important caveat is that the randomness here is conditional on the Santha Vazirani weak source model, the claim that quantum physics can produce randomness, the Bell test, and the validity of the extractors' security analysis."

In fact, the researchers had chosen one of the most stringent applications: a public randomness beacon application for users ranging from governments to military encryption. The U.S. National Institute of Standards and Technology already runs a service where it releases 32 random bits every 60 seconds on its website. They are used in lotteries, to assign jurors, to sample voting machines, and for research.

That said, he wrote, cautioning that this "bit test" state while certified randomness could help security today, it will not help protect encryption information against attacks by future quantum computers.

Better randomness was never the answer to that problem," he wrote. "Inspiring to post quantum algorithms as India recently took its very first steps on this front. (arxiv.org/abs/2008.00001)

and less efficient, it can produce randomness of a higher quality. That is, the work establishes that randomness can be amplified in lab conditions, that too in a device-independent way, that is, even if the device used to generate the randomness is not trusted, the output can be certified to be random.

The team also said their protocol to generate purely random numbers is device-independent – a gold standard in quantum physics that means even if you don't trust the person who built the hardware, even if you don't fully understand how the machine works, you can trust the output to be random.

In physics, it is mathematically impossible for a classical theory to score higher than 2.05 in a bit test. Its scoring 2.23 in the test, then, the researchers' setup proved it was producing random quantum information.

That said, according to the team's security analysis, the protocol has a failure probability of one in a trillion, similar to flipping a coin and getting heads 40 billion times in a row. This is because of the number of tests, 1.3 billion. If they had conducted them more conditionally on the researchers may have reduced the failure rate to one in a superposition of multiple states. It is mathematically impossible to reach 100%.

In their test, the researchers needed about 2.23 bits of information to produce one bit of randomness. And to produce a large amount of data in the end. If their score had been higher, they could have reached the same large amount of randomness with much less data.

While one in a trillion is useful for practical applications, the experiment's "apparatus" can't replace conventional random number generators. First, it's a complex and resource-intensive. Its randomness output is modest compared to commercial systems, around 1,600 bits per second versus 1 billion bits per second. It is also information-wise inefficient in that for every 10 "almost perfectly random" bits it consumes, it produces 1 "verified random" bit.

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But while the new protocol is slower and less efficient, it can produce randomness of a higher quality. That is, the work establishes that randomness can be amplified in lab conditions, that too in a device-independent way, that is, even if the device used to generate the randomness is not trusted, the output can be certified to be random.

The Oman CEPA, a new gateway for India's exports

India and Oman share one of the oldest trading relationships, with commercial and maritime links dating back thousands of years. From the movement of spices, textiles and frankincense to enduring people-to-people ties, trade has long been a cornerstone of this partnership. The India-Oman Comprehensive Economic Partnership Agreement (CEPA), which came into force on June 1, 2026, reaffirms and strengthens this historic relationship by providing a modern framework to deepen trade, investment and economic cooperation for the future.

India is amongst the top suppliers for Oman and bilateral trade has grown from \$8.94 billion in FY2023-24 to \$11.18 billion in FY2025-26, reflecting the growing complementarities between the two economies.

An expansion of India's trade ties

The CEPA comes at a time when India is actively diversifying its trade partnerships and integrating more deeply with global value chains. Following agreements with the United Arab Emirates, Australia, the European Free Trade Association, the United Kingdom, New Zealand and the European Union, the Oman CEPA further strengthens India's presence in a region that is central to its energy security, trade and strategic interests.

The CEPA offers significant opportunities for Indian exporters. Oman has offered duty-free access on 98.08% of its tariff lines, covering 99.38% of India's exports by value. Before the agreement, only 15.33% of India's exports entered Oman at zero duty under the Most Favoured Nation regime. The CEPA, therefore, provides an immediate competitiveness boost across a broad range of sectors. In textiles and apparel, India already commands a 43% share of Oman's woven apparel imports and 31% of knitted apparel imports. The removal of the existing 5% tariff will strengthen the competitiveness of Indian manufacturers against China, which is the other dominant supplier in this market. In the case of chemicals, India already supplies nearly 39% of Oman's inorganic chemical imports, making it



Anant Goenka

President,
The Federation of
Indian Chambers
of Commerce and
Industry (FICCI)

The agreement
boosts exports,
services,
investment and
professional
mobility

one of the leading players in the market. Tariff free access will only amplify this further.

Likewise, engineering goods will also benefit. Oman imports over \$3.7 billion worth of mechanical machinery and \$3.3 billion worth of automobiles annually, where India's market share is only 5% and 2%, respectively. Preferential market access under the CEPA can help Indian exports of engineering goods expand significantly and deepen their presence in Oman's infrastructure, construction and industrial sectors.

In pharmaceuticals, where India holds around 10% market share, the agreement's value lies not in tariff reductions but in regulatory facilitation. Products approved by leading international regulators will benefit from fast-tracked approvals, reducing compliance costs and accelerating market entry. As Oman's pharmaceutical market continues to expand, Indian companies will be well-positioned to capture a larger share. Duty-free access for products such as meat, eggs, honey, butter and processed foods will further strengthen India's already strong position. At the same time, sensitive sectors such as dairy, cereals, edible oils and several agricultural commodities have been kept outside tariff concessions, ensuring that domestic producers remain protected.

Streamlining procedures

Significant trade facilitation measures have been introduced. Oman will accept certificates issued by India's Export Inspection Council (EIC), eliminating duplicative testing and inspections, while also recognising India's organic (NOP) and halal certification systems. There are dedicated sanitary and phytosanitary (SPS) and technical barriers to trade (TBT) provisions, which will enhance regulatory transparency and cooperation, and streamlined customs clearance, including fast-track processing for perishables, which will reduce costs and improve export efficiency.

Another forward-looking aspect of the CEPA is its strong focus on services and professional mobility. Bilateral services trade stood at \$863

million in 2024, with India enjoying a surplus of nearly \$447 million. Yet, India's share in Oman's global services imports remains just over 5%, indicating substantial untapped potential.

Oman has undertaken binding commitments covering professionals in sectors such as accounting, engineering, information technology, health care, education and consulting. Oman also raises the quotas for intra-corporate transferees, facilitating greater mobility of Indian professionals and specialists. Provisions relating to AYUSH and traditional medicine further create opportunities for Indian health care and wellness services in the Gulf.

Strategic location advantage

Beyond trade statistics, the CEPA has a larger strategic significance. Oman occupies a unique position at the crossroads of the Gulf, the Indian Ocean and East Africa. Its ports at Sohar, Duqm and Salalah are emerging as major logistics and industrial hubs. For Indian businesses, Oman can serve not only as a destination market but also as a gateway to the wider Gulf Cooperation Council (GCC) region and East African economies.

The India-Oman CEPA once again demonstrates the evolution of India's trade policy from tariff negotiations to comprehensive economic partnerships encompassing goods, services, investment, mobility and regulatory cooperation. Its benefits will extend from textile clusters in Tamil Nadu and the gems and jewellery industry in Gujarat to engineering hubs in Maharashtra and Punjab, and from pharmaceutical manufacturers in Telangana to seafood exporters in Andhra Pradesh and Kerala.

The real test will now lie in implementation and utilisation. If businesses actively leverage the opportunities created by the agreement, the CEPA can significantly expand India's export footprint, strengthen economic integration with the Gulf, and support the country's broader ambition of becoming a globally competitive manufacturing and services powerhouse. For India, the agreement opens not only the Omani market but also a broader gateway to the Gulf and beyond.

Fifteen FTAs, 27 countries, four challenges

WITH THE India-Oman agreement taking effect on June 1, India now has 15 FTAs covering 27 countries. Another nine agreements with 42 countries are nearing completion. Once finalised, India's FTA partners will total 69 countries and could account for nearly 75 per cent of the country's exports.

As India expands its network of FTAs, four recurring challenges demand attention: Rising trade deficits, low utilisation of FTA benefits by Indian exporters, worsening inverted duty structures, and the relocation of manufacturing to FTA partner countries.

Let's examine how these challenges are shaping India's trade patterns, industrial competitiveness, and economic outcomes.

First, a rising trade deficit. Between 2007-09 (before the FTAs took effect) and 2020-25, India's trade deficit with ASEAN grew by 381 per cent, that with Japan by 318 per cent and that with South Korea by 268 per cent. In comparison, its trade deficit with the rest of the world increased by 142 per cent. Over the past three years, India's average annual trade deficit with ASEAN, Japan and South Korea has reached about \$62 billion.

The newer FTAs are also associated with large trade deficits. In FY2025, India exported \$48.6 billion to the UAE, Australia, Mauritius and EFTA countries, but imported nearly \$100 billion, resulting in a trade deficit of over \$50 billion. As tariff cuts under these agreements deepen, the deficit may increase further. South Asia remains the major exception, where India's trade surplus expanded from \$6.7 billion to \$20 billion during the same period.

The difference between India's tariff structure and those of its FTA partners helps explain why imports often grow faster than exports after FTAs. Most of its FTA partners are already open economies with low tariffs. Average MFN tariffs are close to zero in Sin-

gapore and below 4 per cent in Japan, Australia, Malaysia and the UAE. In contrast, India's trade-weighted MFN tariff is about 12.6 per cent, with rates ranging from zero to 150 per cent. As a result, when India cuts tariffs under an FTA, exporters from partner countries gain a significant price advantage in the Indian market. A 50 per cent tariff reduction, for example, can translate into a major cost advantage over competing suppliers. Indian exporters, however, often gain little additional market access because tariffs in partner countries were already low or zero before the agreement.

The difference becomes even clearer when actual trade flows are examined. Almost all imports into Singapore enter duty-free under MFN rules, while more than 80 per cent do so in Japan and Malaysia. In the EU and the UK, more than half of imports face zero customs duty. In India, however, only about 6 per cent of imports enter duty-free under MFN treatment. As a result, FTAs often give foreign exporters a much bigger advantage in the Indian market than Indian exporters receive abroad.

Second, low utilisation of FTA benefits. The same tariff asymmetry also helps explain why Indian exporters make limited use of FTAs. When MFN tariffs in partner countries are already zero, there is little benefit in exporting under an FTA. Even where MFN tariffs are low — say 1 to 3 per cent — the savings are often too small to justify the costs of complying with rules of origin, certification requirements, and paperwork. As a result, only an estimated 20-30 per cent of India's eligible exports take advantage of FTA preferences. Many small firms prefer to avoid the compliance burden for modest tariff savings.

The incentives are very different for exporters selling to India. Since India's MFN tariffs remain



AJAY
SRIVASTAVA

When it becomes cheaper to manufacture in an ASEAN country and export duty-free to India than to produce in India, investment and jobs tend to move abroad. As a result, FTAs can encourage firms to 'Make in ASEAN, Sell in India'

relatively high, tariff reductions under FTAs can generate substantial savings. As a result, import-side utilisation rates are estimated at 60-70 per cent. Thus, rising imports and low export-side utilisation are not separate issues. Both stem from the same tariff asymmetry between India and its FTA partners.

Third, worsening inverted duty structures. An inverted duty structure arises when duties on raw materials and industrial inputs are higher than those on finished products. While this problem has existed for years, FTAs have made it harder to fix because many finished goods now enter India at low or zero duty from partners such as ASEAN, Japan, South Korea, the UAE and Australia. As a result, Indian manufacturers often pay high duties on imported inputs, especially those sourced from non-FTA countries, while competing against finished products imported duty-free under FTAs.

For example, steel and aluminium attract MFN duties of 7.5-10 per cent, but machinery, industrial equipment and engineering products made from these materials can enter India duty-free under several FTAs. Indian manufacturers, therefore, face higher input costs when competing with tariff-free imported machinery produced with globally priced inputs.

Similar distortions exist in chemicals, plastics, rubber and textiles. Duties on inputs such as caustic soda, soda ash, polypropylene, PVC and SBR raise production costs. At the same time, many finished products in these sectors can be imported at low or zero duty. The result is a tariff structure that protects producers of basic materials but disadvantages downstream manufacturing, making it harder to achieve higher domestic value addition and the goals of Make in India.

Fourth, make in ASEAN, sell in

India. An equally significant consequence of FTAs and resulting inverted duty structures is the growing incentive for firms to manufacture outside India rather than within it. When raw materials and components attract duties in India, but finished products can be imported duty-free from FTA partners, companies may find it more profitable to locate production abroad and export back to the Indian market. In such cases, FTAs effectively encourage offshore manufacturing at the expense of domestic value addition.

ASEAN countries are increasingly becoming manufacturing hubs for supplying the Indian market. Chinese companies have invested heavily in countries such as Vietnam, Thailand and Indonesia. At the same time, some Indian firms have also set up factories and joint ventures there to benefit from lower production costs and duty-free access to India under FTAs. Similar trends can be seen in electronics, steel, chemicals, plastics, consumer goods and engineering products.

When it becomes cheaper to manufacture in an ASEAN country and export duty-free to India than to produce in India, investment and jobs tend to move abroad. As a result, FTAs can encourage firms to 'Make in ASEAN, Sell in India' rather than 'Make in India'.

Unless India's tariffs on industrial inputs are better aligned with its FTA commitments, these agreements may encourage firms to produce abroad rather than in India, weakening domestic manufacturing and supply chains.

The government and industry must work together to address these four challenges so that FTAs strengthen India's manufacturing base instead of encouraging higher imports, overseas production, and loss of industrial capacity.

The writer is founder, GTRI

Why Census 2027 matters for development, democracy and representation

Census 2027, the world's largest count of people within a national geography, is an overdue exercise with far-reaching implications; an accurate and comprehensive count will ensure that no one is left behind and that policies, entitlements, and resources are guided by reliable data

Akshay Raut

Record turnouts in the recently concluded Assembly elections have brought satisfaction to all those who believe in the ultimate say of numbers and counts in a democracy. The only concern amid this celebration is whether everyone who needs to be counted has actually been included. This sense of urgency should equally drive the ongoing Census, the mother of all counts.

The world's largest-ever count of people within a national geography got underway on April 1, with the commencement of the first of the two phases of Census 2027. The first phase will gather nationwide information on the conditions, amenities, and assets of every household. The second will collect critical demographic and socio-economic information, including details on education, migration, fertility, and, of course, caste.

Seriously overdue

The eighth Census since Independence comes after a gap of 15 years instead of the usual 10, owing first to the COVID-19 disruption and later to the Lok Sabha elections. India has certainly been missing an updated demographic database. Public and private planning alike have had to rely either on the 2011 Census or on informed extrapolations from it.

There are other credible surveys, including five rounds of the National Family Health Survey (NFHS). But these largely focus on health and nutrition. The World Bank and various UN agencies have also periodically conducted studies in their areas of interest. The United Nations Population Fund estimates India's population in 2025 at more than 146 crore. The population count in 2011 stood at 121 crore, marking a growth of 17.7% over the 2001 Census. Census 2027 is expected to record an increase of 25-30 crore. That makes the case for an accurate, secure, and comprehensive national census even more urgent.

The purpose of the Census in colonial India was to help the rulers govern more efficiently and sustain the regime. In post-independence India, the Census has served a different purpose: improving governance and speeding up multidimensional and inclusive development. The 1951 Census, expectedly, revealed dismal levels of literacy, poverty, and life expectancy in a country depleted by colonial rule. India in 2026-27 stands on a completely different footing in terms of mobility, connectivity, infrastructure, education, healthcare and other indicators of living standards, which await measurement.

Contested demography

The season of demographic politics is here. Special intensive region (SIR) is a rebuilding of India's electoral rolls; the process has also prompted citizens to ensure that they are counted and that their names remain on the rolls.

The act was played in Parliament in mid-April when women's reservation became a casualty of the fear of 'biased' delimitation for constituencies. New electoral boundaries will ordinarily be anchored in Census data, and the 32%



Census enumerators during a door-to-door survey of digital mapping for the Census at Shashi Garden, in New Delhi, on May 29, 2024. *Kaushik Venka*

reserved seats for women in the legislature will be tied to it. This way, the Census exercise will have elaborate implications in identified areas.

Public policy and service delivery can suffer when residents are not properly counted under various buckets. Digital data collection, being done for the first time, could provide deeper and more segregated identification of issues in localities. Viewed from another perspective, departmental dashboards for social and infrastructural provisions could face reality checks post-Census, against their claims.

Existing reservations for Scheduled Castes and Scheduled Tribes will also have the new Census as a reference. Instead of belligerent battles fought over caste, class, and community interests in the electoral cauldron, getting duly counted along these divisions could be a constructive presentation of the case. Affirmative action for marginalised groups, provided under India's legal and constitutional framework and considered as the fountainhead of social justice, depends on systematic data drawn from the Census.

Roti, kapda, makaan

The Census may appear not to offer any individual profit, but most certainly it does. Access to jobs, food, housing, cooking gas, clean drinking water, safe sanitation, schools, and hospitals are among the expectations citizens have of an elected government.

Accurately identifying needs and deprivations through reliable databases provides the basis for consistent delivery of these welfare measures and public services. Demographic profiles also

inform the Finance Commission's devolution of funds to States and local bodies and even guide private-sector investments in identified areas.

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Guardrails of individual rights

Online self-enumeration, available for 15 days, preceding the survey and already completed by over one crore households, is an additional facilitation, but not mandatory. Self-enumeration blends digital empowerment with the right to privacy and accuracy of personal data. Inclusion is ensured by retaining the manual option and language choices. Census authorities loudly assure that an enumerator will definitely visit your home. The Census Act, 1946, ensures the protection of individual data in the face of a malevolent and interconnected world.

In contemporary elections, population groups look out for targeted benefits, often casting votes in return. The ethical and economic prudence of this bargain is widely questioned. But a good Census delivers the same without moral or fiscal damage. While the government has pitched the Census as an instrument of development, citizens must proactively

grab the chance of getting counted. Census in 2011 had the tagline 'Our Census, Our Future'. In 2027, it is 'Our Census, Our Development'. One could hasten to add, 'my census, my right.'

Everyone's task, everyone's gain

High profile enumeration of the top brass in the national and State capitals carries great symbolism. But the true mandate of the Census is to ensure that no one is left behind and that no facts – or even atrocities – from the remotest hamlet or the most crowded urban slum, regardless of people's circumstances, are excluded from this long-overdue exercise.

Over three million census staffers will have to overcome topographical and social barriers as they go door to door to complete this onerous task.

Although participation in the Census is mandatory, public cooperation must be driven by a heightened awareness of self-interest rather than the apathy that often characterises urban life. *Pragati* and *Vishal* the mascots of Census 2027, are statements not only of the purpose, but of equal role for men and women.

Millions of voters stood in line at poll booths over the past two months because they recognised the importance of their vote, even though voting remains optional. In an ocean of census operations, every individual count matters, just as every single vote matters in a large election.

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Thank you

“The price of success is daily effort”



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